SASKEN

Computer Vision based Fire Detection System

Whitepaper

Abstract

Many fire detection devices have been around for quite some time and they are essential safety devices. This white-paper presents a computer vision based fire detection system. The system takes input from a video camera and on analyzing the input, decides whether fire pixels are present in the input feed or not. To analyze, the system uses color and luminance features of the input images. Based on the decision, the system can raise a fire alarm.

Index Terms - Computer Vision, RGB and YCbCr color spaces, ROI.

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Table of Content

Introduction

The devices that automatically detect fire have been around since 1890 when Francis Upton invented the first smoke alarm [1] [2]. With further technological advances in the mid 1960s, smoke detectors started being used in buildings all over the world and became an essential safety device [3]. But the devices like smoke detectors have some serious limitations which make them inefficient in many important situations. For example smoke detectors, the most common fire detection device, works properly in small enclosed spaces like homes or offices. However, in large open spaces like warehouses, theaters, verandas, these devices are inefficient as they require sufficient levels of smoke build-up to turn them on. A more modern approach like electronic fire detectors usually depend upon heat and pressure sensors. They also suffer from the same limitation as smoke detectors and require certain amount of heat and pressure built up to set them off which is an impractical approach in case of open spaces. Also in case of forest fires, these can't be detected using smoke and electronic detectors. Due to rapid developments in digital camera technology and advancements in computer-vision based methods, more and more computer-vision based fire detection systems are being introduced. Video-based fire detection do not suffer from space limitations that smoke and heat detection does. Video-cameras can detect and pin-point location of fire as soon as the fire starts. This allows the fire to be dealt with way before it gets out of control. In further advancements to the system, these video-based fire monitoring systems can even be placed on robots and UAVs to detect fire in remote forest areas.

Methodology

\As fire is a visual phenomenon, a multi-feature-based approach to detect fire is discussed in [1] and [4]. Fire has many distinctive features like color, motion, etc and the discussed algorithm extracts these features to classify whether the pixels under consideration are of fire or not. As fire is not a stationary process and to reduce the computational load, the first step in the algorithm is to detect in which region of the image under consideration captured by the video camera, motion is happening. This is done by frame differencing the n^{th} frame (i.e. the frame under consideration) of the video with $(n-1)^{th}$ frame and the obtained resultant pixels are called Region of Interest (ROI) in the image. These ROI pixels are then fed to Fire Pixel Classifier (FPC). The FPC consists of various rules and if the ROI pixels pass all these rules, then it is said that the pixels under consideration belong to fire. Upon detection of fire pixels, the system can raise an alarm. The above procedure can be well understood with the help of the following flow chart shown in **Fig. 1:**

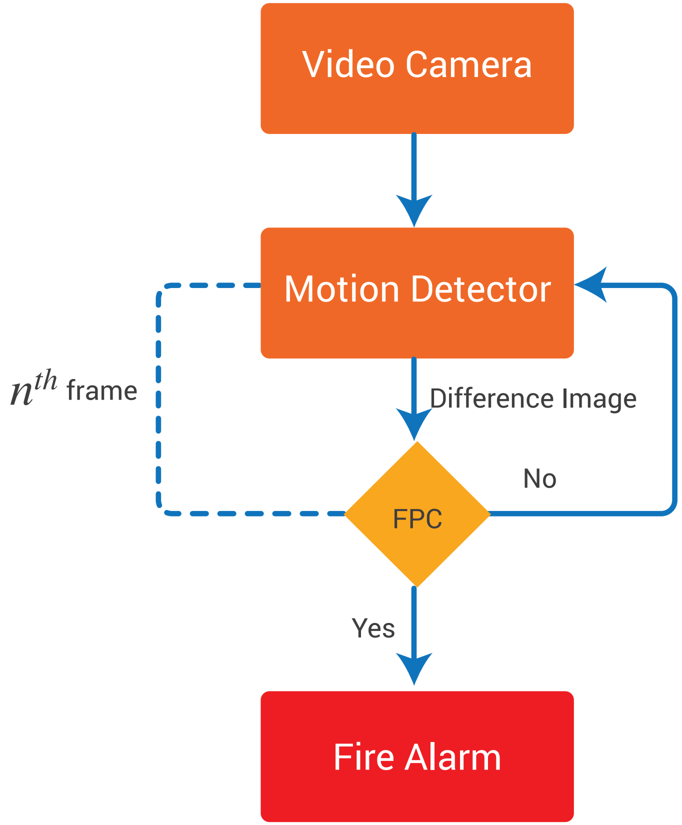

Fig. 1 Flow chart of the proposed System

The various components of the system are as:

A. Motion Detector:

As flames of fire often flicker and jump, the first step in the algorithm is to detect in which region of the image under consideration, motion is happening. This is done by differencing the subsequent frames of the captured video based on their intensity values. Let n^{th} frame of the video be the frame under consideration. As this frame is a color image and the color image consists of 3 channels (i.e. R, G, B channels), motion is detected by differencing the intensity values of R, G, B channels of n th frame with $(n-1)^{th}$ frame. The resultant pixels obtained from this differencing operation constitute the ROI and further processing will be done on these pixels. **Fig. 2** shows the two subsequent frames of captured video and **Fig. 3** shows the difference image obtained after applying differencing operation. The three channels R, B and G in the difference image are merged and converted to gray-scale for better visualization of result. The block diagram of motion detector is as shown in **Fig. 4**.

Fig. 2 Sample subsequent frames of captured video

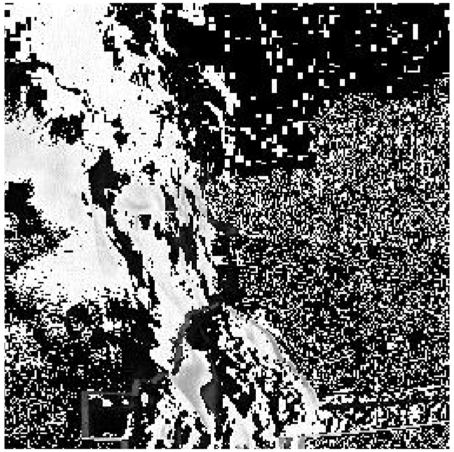

Fig. 3 Result obtained after differencing operation

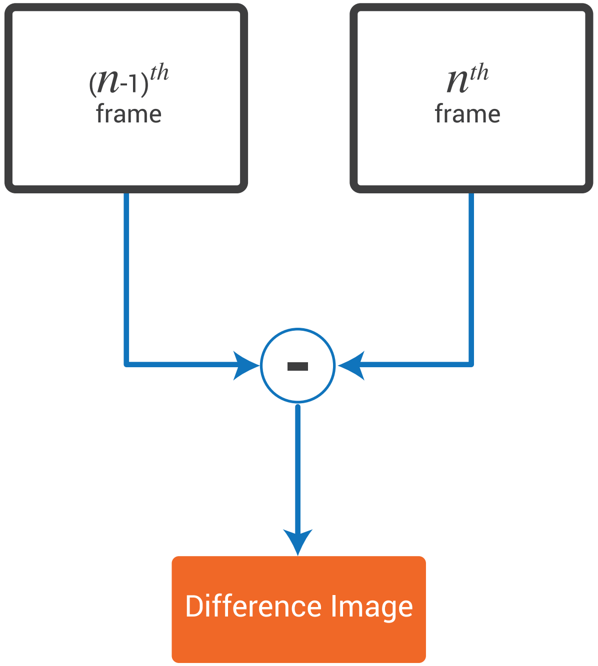

Fig. 4 Motion detector block diagram

B. Fire Pixel Classifier (FPC):

In FPC, rule based color model approach is followed. As fire pixels have color and luminance properties, the obtained ROI is processed in RGB and YCbCr color spaces. To be identified as fire pixels by FPC, the ROI pixels have to satisfy certain rules. If ROI pixels satisfy all these rules, then they are classified by FPC as fire pixels. If not, then those set of pixels are dropped and the process repeats for new ROI. This process can be visualized with the help of the flow chart as shown in **Fig. 5:**

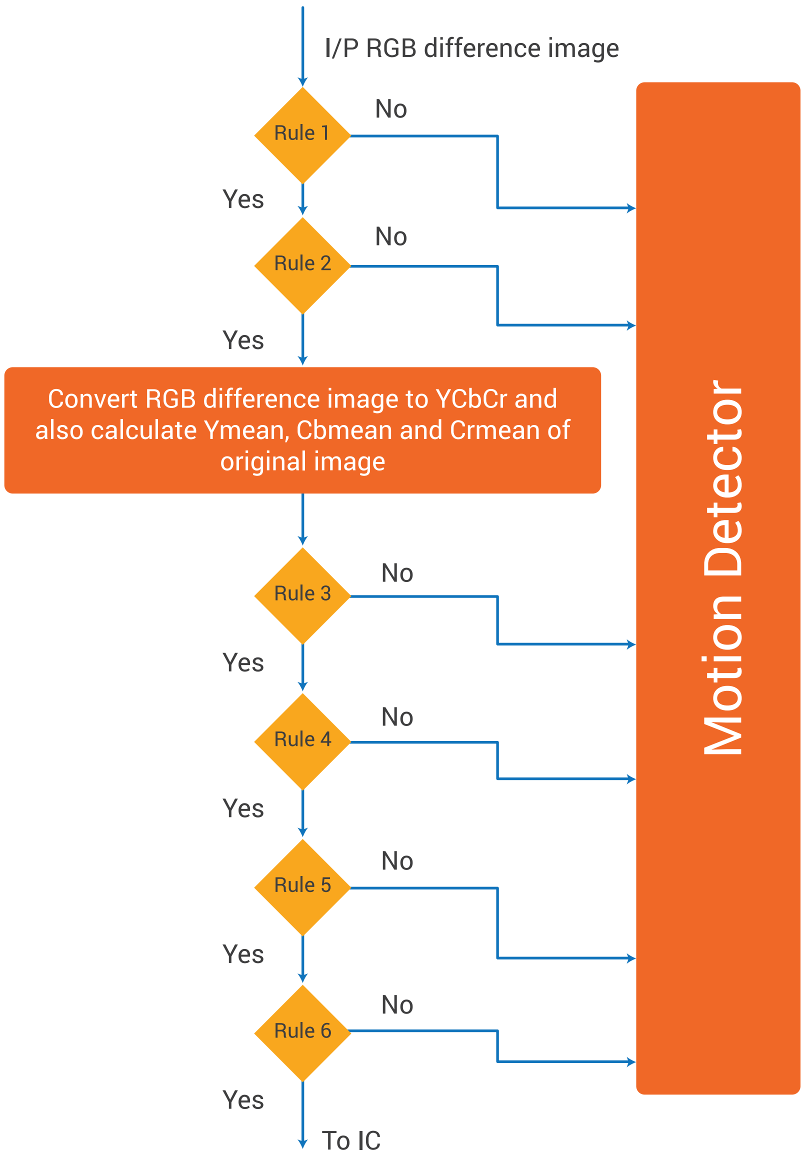

Fig. 5 Flow chart of Fire Pixel Classifier (FPC)

The various rules that ROI pixels have to satisfy are:

Rule I: For fire pixels in case of RGB color space, R channel has higher intensity values than G channel, and G channel has higher intensity values than B channel. So, a pixel value at location (x, y) in the difference image is said to be fire pixel if $R(x, y) > G(x, y) > B(x, y)$.

Rule II: **Fig. 6** shows highlighted fire pixels region in the original image and **Fig. 7** shows the histograms of R, G and B channels of the highlighted fire pixels:

Fig. 6 Original RGB image on the left and image with highlighted fire pixels on the right.

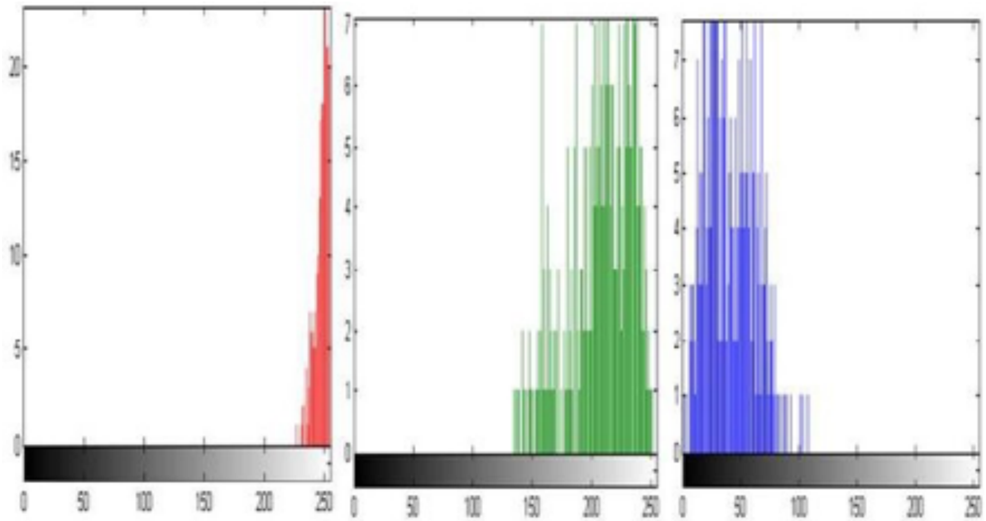

Fig. 7 Histograms of R, G and B channels of highlighted fire pixels

The logical explanation of this behavior is that image of fire is composed of yellow color with traces of red color in it. Yellow color in turn is a combination of green and red color. So, the histogram of fire image will have a number of red components in high tonal range, green components from medium to high tonal range and the blue components will lie in low tonal ranges.

Based on the above observations, a threshold Rth for R-channel, Gth for G-channel and Bth for B-channel can be set such that for a pixel at location (x, y) in difference image is said to be fire pixel if $R(x, y) > R_{th}$ and $G(x, y) > G_{th}$ and $B(x, y) < B_{th}$.

Images of RGB color space can be converted to YCbCr color space. For the nth frame of video, the mean

values of Y, Cb and Cr channels denoted as Y_{mean}, Cb_{mean}, Cr_{mean} are calculated.

Rules III and IV: The intensity of fire region in Y and Cr channels is more than that of Cb channel. So, the pixel at location (x, y) in the difference image is said to be fire pixel if $Y(x, y) \geq Cb(x, y)$ and $Cr(x, y) > Cb(x, y)$.

Rule V: As fire region in an image is made up of yellow and red color components, Cr of fire region pixels will be greater than mean Cr of whole image and Cb of fire region pixels will be less than Cb of whole image. Also as fire region in an image is of high intensity, Y of fire region pixels will be greater than mean Y of whole image.

Rule VI: **Fig. 8** shows the RGB image and its YCbCr converted image and **Fig. 9** shows histograms of Cb and Cr channels of the fire region in YCbCr image:

Fig. 8 Original RGB image on the left and RGB to YCbCr converted original image on the right

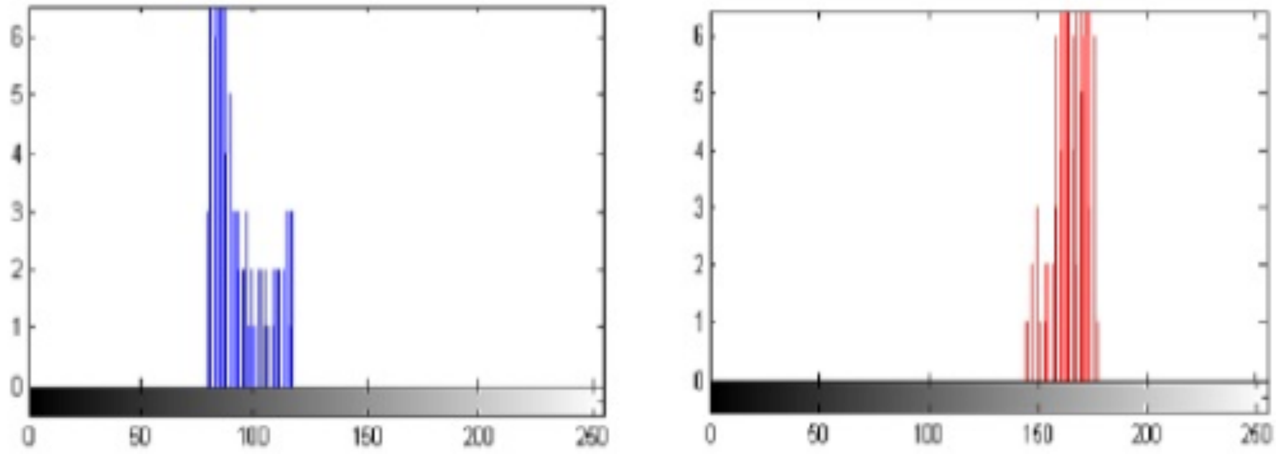

Fig. 9 Histograms of fire region of YCbCr image. Histogram of Cb channel on the left and Cr channel on the right.

Hence, the thresholds Cb_{th} and Cr_{th} for the channels Cb and Cr can be set such that for the pixel at location (x, y) in the difference image is said to be fire pixel if $Cb(x, y) \leq Cb_{th}$ and $Cr(x, y) \geq Cr_{th}$.

Performance Evaluation

Based on above discussed methodology, the authors in [4] used classification error matrix for performance evaluation. For this, they collected two sets of images. One set composed of images that consisted of fire. The fire set consisted of 200 images, with diversity in fire color and environmental illuminations. The other set didn't contain fire, but contained fire color regions like sun, flowers, etc.

The following condition was used to declare a fire region: if the model achieves to detect at-least 10 pixels of fire in ROI, only then it is assumed that the image contains fire region and fire alarm is raised.

		Actual data: Fire	Actual data: No Fire	
Classified data	Fire	198	28	226
	No Fire	2	172	174
		200	200	

Fig. 10 Classification Error Matrix

Fig. 10 shows the classification error matrix obtained by the authors in [4]:

With above methodology, the authors in [4] were able to achieve classification accuracy of 92.5%.

Conclusion

The paper discussed the existing algorithms for fire detection through vision sensors based on the color-space approach. RGB and YCbCr color spaces are used by authors in [4] and [1] to obtain the classification accuracy of 92.5%.

The further advancement to the system may involve machine learning to achieve high classification accuracy and to decrease false-alarm rates.

References

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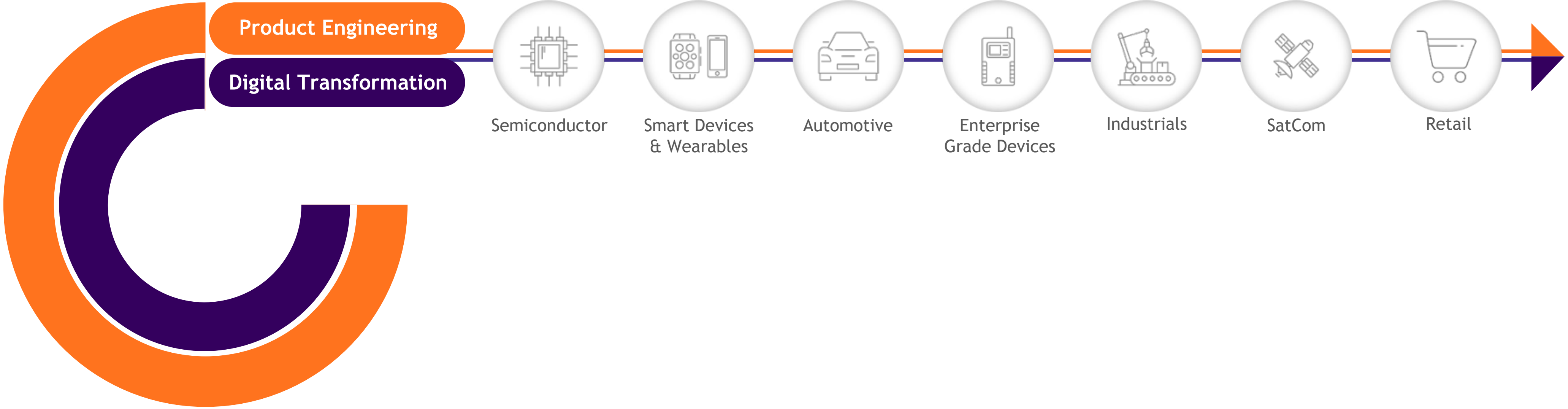
About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Smart Devices & Wearables, Enterprise Grade Devices, Satcom, and Retail industries. With over 27 years in Product Engineering and Digital Transformation and 70 patents, Sasken has transformed the businesses of over a 100 Fortune 500 companies, powering over a billion devices through its services and IP.



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June 2017